Biochemistry Cume – Protein Structure and Folding
January 17, 2007

1. Calculate the net charge on the peptide, Arg-Ile-Val-Glu-Arg, at pH 4.8 and 11.8. In addition, find the pH at which the peptide is expected to be electrically neutral. Assume that there are no intramolecular charge interactions in the peptide at any pH. (20 points)

2. Which polypeptide would you expect to be more compact at neutral pH, polyalanine or polylysine? Which of the two homopolypeptides would show the greatest dependence of shape on pH? Why? (20 points)

3. In a study, a 13 amino acid helical peptide was shown to undergo salt-dependent conformational transitions. Depending on the salt concentration in the solvent, the peptide exists either as a α-helix or a 3_{10} helix or a π-helix.
   - **A.** List at least three distinguishing features of α-helix, 3_{10} helix and, π-helix.
   - **B.** How many complete turns of a π-helix can be accommodated in the 13 amino acid peptide?
   - **C.** Which of the three helical conformations is energetically the most <u>un</u>stable (least stable)? Justify your answer. (20 points)

4. A protein contains six cysteines and all of them are engaged in disulfide bonds. How many different combinations of disulfide pairings can be formed from six cysteine residues? (10 points)

5. **A.** Briefly describe the experiment which led to the conclusion that all the information for a protein to fold to its unique native conformation is present in the primary amino acid sequence.
   B. Urea-induced equilibrium unfolding experiment was performed on a protein at 25° C. Calculate the free energy change of unfolding (in kcal/mol.) of the protein in 2 M urea, if 20% of the protein molecules exist in the denatured state(s) (in 2M urea). (20 points)

6. **A.** What do ϕ and ψ angles in a Ramachandran plot denote?
   B. How is the Ramachandran plot useful in judging the quality of three-dimensional structures of proteins determined by NMR and X-ray methods? (10 points)

Information that can be useful.

Charged group	pK_a
α-carboxyl group	2.5
β-carboxyl group	3.6
γ-carboxyl group	4.8
α-amino group	8.9
ε-amino group	9.4
guanidino group	12.8

Universal Gas constant - 1.9872 $cal.mol.^{-1}.K^{-1}$